

Jonathan Barnes | Applied Data Scientist

Minneapolis, Minnesota | <https://jonathanmbarnes.github.io> | Jonathanmbarnes@outlook.com

PROFESSIONAL EXPERIENCE

University of Minnesota, Minneapolis, MN

Analytics Engineer | Center for Public Health Systems

(May 2025 - Present)

- Architected a production analytical data platform for all HHS awards, consolidating 400 inconsistent source fields into 4M records and 26 standardized variables enabling global funding analysis that was previously infeasible at scale.
- Designed a fault-tolerant repeatable ingestion and orchestration layer backed by DuckDB that replaced unreliable government APIs, undocumented rate limits, and pagination constraints, and serves as a persistent analytical store that supports refresh and future ad-hoc querying through custom built interface for non-technical users.
- Engineered a memory-efficient columnar normalization pipeline using Polars that processes 60 GB of raw heterogeneous federal data end-to-end under 2 minutes with under 12 GB of working memory, reducing the memory requirement by over 80% and eliminating the need for high-memory or distributed infrastructure.
- Imposed analytical coherence on inconsistent records by reconciling schema drift, missingness, and contradictory values by coalescing across 4–8 fields per variable and using external reference tables increase accuracy and validity.
- Built a custom Polars-to-Tableau Hyper export layer to reduce analytic file size by over 90%, significantly improving enterprise dashboard performance while preserving full granularity.
- Integrated PHIG grant data with HHS awards to power Tableau dashboards and cross-program visual analysis.

Graduate Teaching Assistant | Division of Biostatistics and Health Data Science

(Aug 2024 - May 2025)

- Taught and evaluated applied statistical analysis in R, improving student code quality and statistical reasoning.

Graduate Research Assistant | Medical School

(Oct 2023 - May 2024)

- Designed custom plasma biomarker phenotypes and gene expression features to enable survival modeling.
- Applied Cox proportional hazards regression to model time to cognitive decline using biomarker phenotypes.

University of St. Thomas, St. Paul and Minneapolis, MN

ITS Support Technician Intern & Senior Student Technician | ITS Support Services

(Aug 2021 - Aug 2023)

- Resolved or consulted on 4,500+ support tickets across hardware, software, networking, and security, serving as the senior escalation layer between the service desk and support teams while shaping the design of standardized process.
- Designed and deployed a Python-based internal platform that standardized high-variance service desk workflows, replacing a legacy tool and becoming the primary interface used by technicians during live support.
- Built guarded abstraction layers over operating system, directory, and network queries to provide consistent, repeatable access to diagnostic and identity information without exposing raw command-line complexity.
- Mentored junior student technicians to independently handle complex service desk operations and consultations.
- Imaged and administered software on 1,500+ replacement computers and leveraged retired hardware to develop an in-house virtual lab replacing AWS EC2 instances and reducing infrastructure costs.
- ITS Outstanding Student Award.

EDUCATION

University of Minnesota School of Public Health, Minneapolis, MN

Master of Public Health - Public Health Data Science, GPA 3.73/4.00

(Expected Jan 2026)

- Capstone – Data Platform Evaluating Health and Human Services Federal Spending and Systems.
- President's Student Leadership & Service Award Nominee (2025)

University of St. Thomas, St. Paul, MN

Bachelor of Science - Data Analytics (Biology Domain), Completed while employed full-time

- Minors - Biology (Biochemistry & Computational Biology courses), Applied Statistics.
- Capstone - Development of Obesity and Nutritional Trends Over Time in the United States Using Combined NHANES.

SKILLS

Languages: R, Python, SQL, Bash.

Tools: Shiny, APIs, Git, ArcGIS, Tableau, CI/CD, Microsoft Azure, Docker.

Competencies: Data Visualization, Data Communication, ETL Pipelines, Geospatial Visualization, Longitudinal Data Analysis, Machine Learning, Artificial Intelligence, Relational Databases, Columnar Data Architectures, System Design, Experimental Design, Data Acquisition, Consulting, Health Informatics, Electronic Health Records, HIPAA, PHI, SNOMED, ICD-10.

Statistical Methods: Dimensionality Reduction, Supervised Learning, Unsupervised Learning, Regression, Regularization, Tree Models, Mixed Effects Models, Survival Analysis, Structural Equation Modeling, Natural Language Processing.

Supporting Work

Public Health Federal Funding Evaluation Platform & Dashboard (Grant Supported)

- Developed a modular ingestion and modeling pipeline to process millions of federal public health funding records, spanning prime awards, sub-awards, and derived transactions across inconsistent reporting structures.
- Reconciled fragmented and contradictory HHS data by implementing identity modeling and rules-based normalization to link awards, recipients, and health departments across divergent hierarchies, identifiers, and reporting systems.
- Architected columnar storage and transformation layers using DuckDB and Parquet, reducing data footprint by 90%+ while preserving transaction-level performance, and materialized normalized models into analysis-ready datasets and custom Tableau Hyper extracts for scalable, reproducible analytics.
- Built a GUI-based orchestration layer to manage pipeline execution, validation, and controlled data refresh, supporting non-technical users in generating analysis-ready datasets and dashboards on funding flows, geography, and program-level allocation.

Methods & Tools: Python, Polars, SQL, DuckDB, Apache Parquet, Tableau (Hyper), Data Modeling, Schema Design, Entity Resolution, Data Normalization, Data Pipelines, Data Warehousing, Data Validation, Health Informatics.

Tech Desk Tools Python

- Designed and deployed a user-facing Python application that serves as the primary interface for Tech Desk workflows, enabling technicians to initiate identity lookup, diagnostics, and institutional tool access through a unified system.
- Implemented identity derivation and system automation logic that abstracts OS and directory complexity into stable, technician-ready outputs, minimizing manual troubleshooting and workflow variability.
- Integrated OS-level commands and directory services with privilege-aware, role-based execution paths to ensure consistent and secure behavior across diverse account, permission, and system states.
- Adopted as the standard Tech Desk application, supporting continuous daily operations since early 2023.
- <https://jonathanmbarnes.github.io/Sections/Portfolio/PortProjects/TDTPython.html>

Methods & Tools: Python, PowerShell, Bash, App Development, Active Directory, Azure, Systems Automation, Workflow Orchestration, Error Handling & Defensive Programming, Technical Writing, Packaging and Deployment.

Mapping Misery – Predicting Poor Physical and Mental Health Days in the U.S.

- Developed predictive models for poor physical and mental health days in 3,000+ U.S. counties using R.
- Engineered clusters based on 10 years of trends across 15 metrics using Gaussian Mixture Models (GMM) by transforming variables into 8 level categories with NAs for robust Multiple Correspondence Analysis (MCA).
- Achieved an R^2 of 0.844 (RMSE: 0.263) for physical health days with Elastic Net Regression.
- <https://jonathanmbarnes.github.io/Sections/Portfolio/PortProjects/MappingMisery.html>

Methods & Tools: Data Pipelines, PCA, MCA, GMM Clustering, Feature Engineering, Longitudinal Data Analysis, Elastic Net Regression, Predictive Modeling, R, Geospatial Visualization, GIS, Data Visualization.

Predicting Child Mortality and Cesarean Delivery Outcomes with Generalized Linear Mixed Models.

- Built Generalized Linear Mixed Models (GLMM) to investigate factors affecting child mortality and cesarean delivery.
- Determined a 1,124.5% increase in the odds of cesarean delivery for hospital births (95% CI: 618.7% - 1,986.3%, $p < 0.001$) — a result consistent with logical expectations.
- https://jonathanmbarnes.github.io/Sections/Portfolio/PortProjects/Child_Mortality.html

Skills & Tools: GLMM, GEE, R, Data Cleaning, Data Pipelines, Data Visualization, Feature Engineering, Collaboration.

Applied Biostatistics Introduction Modules (Grant Supported)

- Designed applied biostatistics modules translating into intuitive frameworks with supporting github hosted applications
- <https://jonathanmbarnes.github.io/SampleSizeApp/>
- <https://jonathanmbarnes.github.io/CentralLimitApp/>

Methods & Tools: Biostatistics, Statistical Reasoning, Probability, Curriculum Design, Technical Writing, R, Shiny, Data Pipelines, Dashboards, Data Visualization, Data Communication, JavaScript.

Data Inspection Dashboard

- Developed an interactive data inspection tool that automates EDA, data validation, missingness handling, imputation, and statistical summaries for common public health datasets.
- <https://jonathanbarnes.shinyapps.io/DescriptiveApp/>

Methods & Tools: R, Shiny, Data Pipelines, Dashboards, Data Visualization, Data Communication, JavaScript.